

42. Let R_1 and R_2 be the "divides" and "is a multiple of" relations on the set of all positive integers, respectively.

That is, $R_1 = \{(a, b) \mid a \text{ divides } b\}$ and $R_2 = \{(a, b) \mid a \text{ is a multiple of } b\}$. Find

a) $R_1 \cup R_2$.

b) $R_1 \cap R_2$.

c) $R_1 - R_2$.

d) $R_2 - R_1$.

e) $R_1 \oplus R_2$.

49. How many relations are there on a set with n elements that are

a) symmetric?

b) antisymmetric?

c) asymmetric?

d) irreflexive?

e) reflexive and symmetric?

f) neither reflexive nor irreflexive?

50. How many transitive relations are there on a set with n elements if

a) $n = 1$?

b) $n = 2$?

c) $n = 3$?